

Silei Ren

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EDUCATION

Cornell University Ph.D. Candidate in Computer Science

Ithaca, NY, 2020-2026

- I design secure-by-construction programming languages using information-flow control (IFC) type systems, build secure compilers, and formally verify compiler correctness. Advisor: Andrew C. Myers.

Brown University B.S. in Math & Computer Science

Providence, RI, 2015-2019

- GPA: 3.8/4.0
- CS Department Senior Prize (top 5%)

FIRST AUTHOR PUBLICATIONS

- **S. Ren**, S. Kanniwadi, H. Chen, and A. C. Myers. "Type-Confusion Security" arXiv.
- **S. Ren**, C. Acay, and A. C. Myers. "An Algebraic Approach to Asymmetric Delegation and Polymorphic Label Inference." ESORICS 2025.
- M. Barbaraci*, S. Ma*, H. Malvai*, M. Mouallem*, **S. Ren***, Y. Sato*, S. Yang*, F. Zhang*. "DeadDrop: Responsible Disclosure of Smart Contract Bugs" arXiv.
- E. M. Kornaropoulos*, **S. Ren***, and R. Tamassia. "The Price of Tailoring the Index to Your Data: Poisoning Attacks on Learned Index Structures." SIGMOD 2022. (*Equal contribution)

TECHNICAL SKILLS

Programming Languages: Java, Kotlin, OCaml, Python, C, C#, Matlab, C++, Rocq, X86, Go, Bash, Solidity.

Tools: Git, LaTeX, Docker, Vim, TEE, PyTorch, Phala, GraphQL, LLVM, MLIR, TileIR.

Areas: Formal Methods, Compilers, Security, Cryptography, Systems, Machine Learning, Logics, Algebra.

RESEARCH/PROFESSIONAL EXPERIENCE

Type Confusion Security (Java, Solidity)

Ithaca, NY, 2025-Present

- Spearheaded the first formal framework for Confused Deputy Attacks and Reentrancy Attacks.
- Designed novel language-based mitigations and formally verified the correctness of the mechanisms.
- Mentored Master's students to implement the defense in SCIF, a smart contract compiler.

DeadDrop (Rust/Docker)

Ithaca, NY, 2025-Present

- Co-designed DeadDrop, an anonymous bug bounty system using cryptography and trusted enclaves.
- Formulated bug specification abstraction and implemented automated bug simulation using Foundry.
- Benchmarked bug simulation module in an Intel TDX enclave served by the Phala Cloud.

Formalizing Memory Sharing in Trusted Execution Environment (TEE)

Berkeley, CA, 2023

- Formalized a core calculus for distributed systems with TEEs and symbolic cryptography primitives.
- Proved information-flow security and resistance against side channel attacks and replay attacks.

Viaduct Compiler (Kotlin)

Ithaca, NY, 2022-2025

- Developed an algebraic type system that unified existing theories of information flow control.
- Implemented the type system in Viaduct, a compiler translating IFC programs into crypto protocols.
- Designed and implemented the first security-type inference algorithm that supports heterogeneous trust delegation, global type inference, and bounded IFC-type polymorphism.

Poisoning Attack on ML Indexing (Python)

Providence, RI, 2018-2021

- Designed the first poisoning attack algorithm on ML-based database indexing structures.
- Implemented the attack algorithm and benchmarked it against traditional indexing structures.
- Demonstrated 3000x efficiency decrease for less than 10% data poisoning, proving attack feasibility.

Microsoft SWE Summer Intern (C#)

Sunnyvale, CA, 2017

- Designed and prototyped a proxy web service that translates between OData and GraphQL queries.
- Presented the proxy service to the group, delivered a live demonstration and earned a return offer.

Teaching Assistant (Bash/Docker)

2016-2026

- Led the teaching assistants for 7 courses, including functional programming, systems, machine learning, compilers, cryptography, and advanced programming languages.
- Developed virtualized testing environments for student compilers written in different languages.
- Managed TA staff meetings, grading sessions, student review sessions and gave lectures.

Stack Machine Compiler (LLVM/MLIR)

2026

- Implemented an end-to-end compiler of a simple stack machine using LLVM/MLIR.